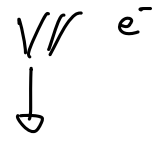
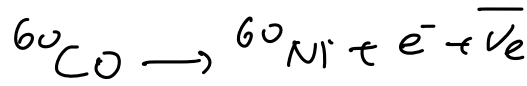


CP Violation

WU experiment :



$\Rightarrow$ exp. proof of $\cancel{P}$

in weak interactions

$$\mathcal{L}_W \sim \gamma^\mu (1 - \gamma^5)$$

$$\mathcal{L}_{EM} \sim \gamma^\mu$$

depends on $\vec{B}$ or orientation of spin of ^{60}Co

Goldhaber experiment

$$\nu \xrightarrow{\mathcal{C}} \bar{\nu} \xrightarrow{\mathcal{P}} \nu \quad \Rightarrow \quad \cancel{\mathcal{CP}} \text{ in weak interactions}$$

$\mathcal{C}$: particle $\rightarrow$ anti-particle

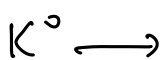
$\mathcal{P}$: $\vec{p} \rightarrow -\vec{p}$

$$\nu \xrightarrow{\mathcal{C}} \bar{\nu} \xrightarrow{\mathcal{P}} \nu \quad \Rightarrow \quad \mathcal{CP} \text{ conserved in this experiment}$$

$\mathcal{CP}$ conserved in this experiment

	$\mathcal{P}$	$\mathcal{C}$	$\mathcal{CP}$
QED	✓	✓	✓
QCD	✓	✓	✓
weak	✗	✗	?

$\cancel{CP}$ in weak interactions: 1964 in K meson decays.



$$\pi^+ \pi^-$$

$$\pi^+ \pi^- \pi^0$$

$$\pi^0 \pi^0$$

possible CP eigenstates

$\pi^+ \pi^-$:

$$CP |\pi^+ \pi^- \rangle = \alpha |\pi^+ \pi^- \rangle$$

$$C |\pi^+ \pi^- \rangle = (-1)^{L+S} |\pi^+ \pi^- \rangle$$

S : total spin of $\pi^+ \pi^-$ $S=0$ because $S_\pi=0$

L : total ang. mom of $\pi^+ \pi^-$ system

$$P |\pi^+ \pi^- \rangle = \underbrace{(-1)(-1)}_{\text{intrinsic parity of } \pi^+/\pi^-} (-1)^L |\pi^+ \pi^- \rangle$$

$$CP |\pi^+ \pi^- \rangle = (-1)^{2L} (-1)^S = +1$$

$$CP |\pi^+ \pi^- \pi^0 \rangle = (+1) \underbrace{(-1)}_{P_{\pi^0}} \underbrace{(+1)}_{C_{\pi^0}=+1}$$

$\pi^0 \rightarrow \gamma \gamma$ EM C conserved.

$$C_{\pi^0} = (C_\gamma)^2 = +1$$

$$CP |\pi^+ \pi^- \pi^0 \rangle = (-1) |\pi^+ \pi^- \pi^0 \rangle$$

$$K^0 \begin{array}{l} \longrightarrow \pi^+ \pi^- \quad CP = +1 \\ \longrightarrow \pi^+ \pi^- \pi^0 \quad CP = -1 \end{array} \quad \text{2/1000 times.}$$

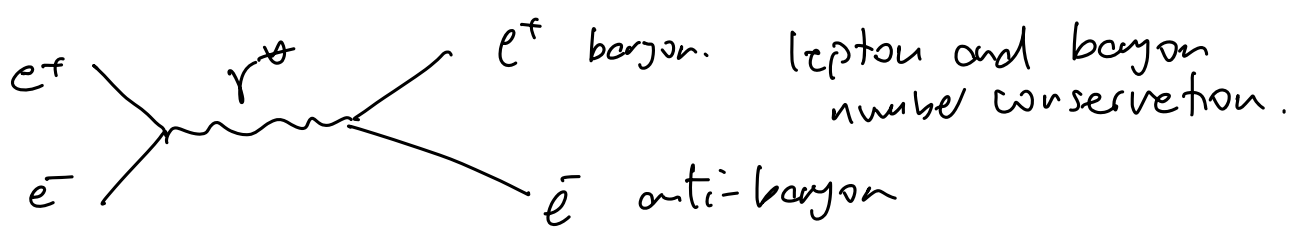
$\Rightarrow$ ~~C~~ in weak interactions.

$\mathbb{T}$: time reversal. $t \rightarrow -t$ $\psi(\vec{x}, t) \rightarrow \psi(\vec{x}, -t)$

theorem in QFT: CPT is a good symmetry

~~C~~ $\Rightarrow$ ~~T~~ in weak interactions

~~C~~ $\Rightarrow$ matter and anti-matter behave differently.



Experimental test:

$$\begin{aligned}
 & \left. \begin{aligned}
 A &: a+b \rightarrow c+d & a \rightarrow b+c \\
 \bar{A} &: \bar{a}+\bar{b} \rightarrow \bar{c}+\bar{d} & \bar{a} \rightarrow \bar{b}+\bar{c}
 \end{aligned} \right\} \text{amplitudes for these processes.}
 \end{aligned}$$

$$\text{if } \not\propto \Rightarrow |A| \neq |\bar{A}| \quad \text{direct } \not\propto$$

we can measure $\#p$, $\#\bar{p}$, $\#e^-$, $\#e^+$, $\#ne$, $\#\bar{ne}$
 $\Rightarrow$ significant asymm.

$$\frac{N_{\text{baryons}}}{N_{\text{protons}}} \sim 10^{-18} \qquad \frac{N_{\overline{\text{baryons}}}}{N_{\text{baryons}}} \sim 10^{-4} - 10^{-6}$$

1967 Sakharov proposal to explain the asymmetry.

- baryon $\#$ violation
- non-equilibrium during evolution
- C and CP violation in interactions

1964: first exp. evidence of $\not\propto$ in weak

$\Rightarrow$ is $\not\propto$ in weak interactions as measured today sufficient to explain our universe?

$\Rightarrow$ study of $\not\propto$ in weak interaction

Types of CP violation :

1) Direct CP violation

$$A : i \rightarrow f \quad i, f: \text{any particle(s)}$$

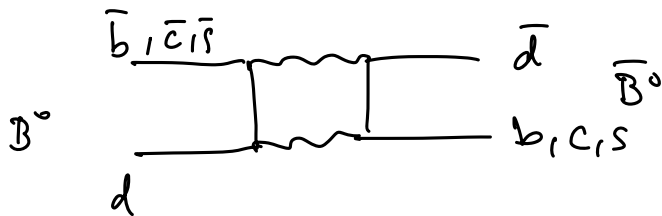
$$\bar{A} : \bar{i} \rightarrow \bar{f}$$

$$\#(i \rightarrow f) \neq \#(\bar{i} \rightarrow \bar{f}) \Rightarrow |A| \neq |\bar{A}|$$

$$B^+ \rightarrow \bar{D}^0 K^+$$

$$\bar{B} \rightarrow D^0 K^-$$

2) CP violation in ^{oscillation} mixing of neutral mesons.



$$B^0 \longleftrightarrow \bar{B}^0$$

$$D^0 \longleftrightarrow \bar{D}^0$$

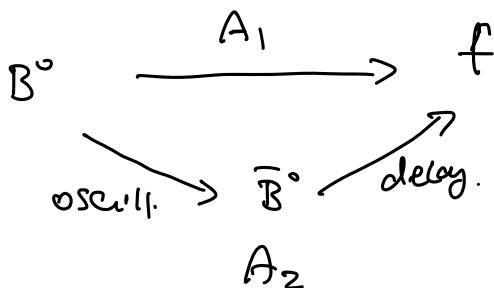
$$K^0 \longleftrightarrow \bar{K}^0$$

$$A_{CP} = \frac{N(B^0 \rightarrow \bar{B}^0) - N(\bar{B}^0 \rightarrow B^0)}{N(B^0 \rightarrow \bar{B}^0) + N(\bar{B}^0 \rightarrow B^0)} = \begin{cases} \emptyset & \text{if CP conv.} \\ \neq \emptyset & \text{if CP} \end{cases}$$

so far compatible w/ $\emptyset$ within 10^{-4}

$$A_{CP}^{mix} \leq 10^{-4}$$

3)



$$A_1 : A(B^0 \rightarrow f) \text{ decay.}$$

$$A_2 : A(B^0 \rightarrow \bar{B}^0 \rightarrow f) \text{ oscillation + } \bar{B}^0 \text{ decay.}$$

$$A_{tot}(B^0 \rightarrow f) = A_1 + A_2 \quad \#(B^0 \rightarrow f) \propto |A_1 + A_2|^2$$

$$\begin{array}{ccc}
 B^0 & \xrightarrow{CP} & \bar{B}^0 \\
 f & \xrightarrow{CP} & \bar{f}
 \end{array}
 \Rightarrow f \equiv \bar{f}$$

f has to be an eigenstate of CP

$$f = \pi^+ \pi^-, \pi^0 \pi^0, K^0 \bar{K}^0, D^0 \bar{D}^0, D^+ D^-$$

$$\#(B^0 \rightarrow f) \propto |A_{tot}|^2 = |A_1|^2 + |A_2|^2 + A_1^\dagger A_2 + A_2^\dagger A_1$$

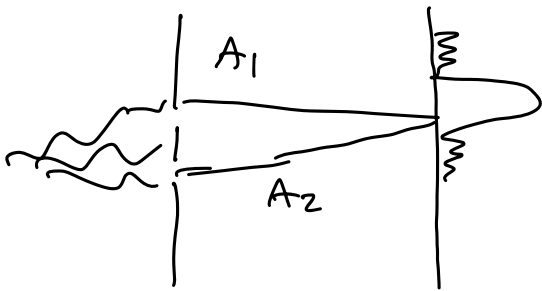
$$A_1 \equiv |A_1| \text{ real } \#$$

$$A_2 = |A_2| e^{i\alpha}$$

$$A_{tot} = A_1 + A_2 e^{i\alpha}$$

$$\bar{A}_{tot} = A_1 + A_2 e^{-i\alpha}$$

} presence of
some complex phase
necessary for
intef. effects.



In CKM matrix

there has to be a complex
phase with at least

3 families

